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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	
Student Name	
Roll Number	
Branch	
Course Outcome	

Case Study Topic: "HealthNet – Intelligent Healthcare System"

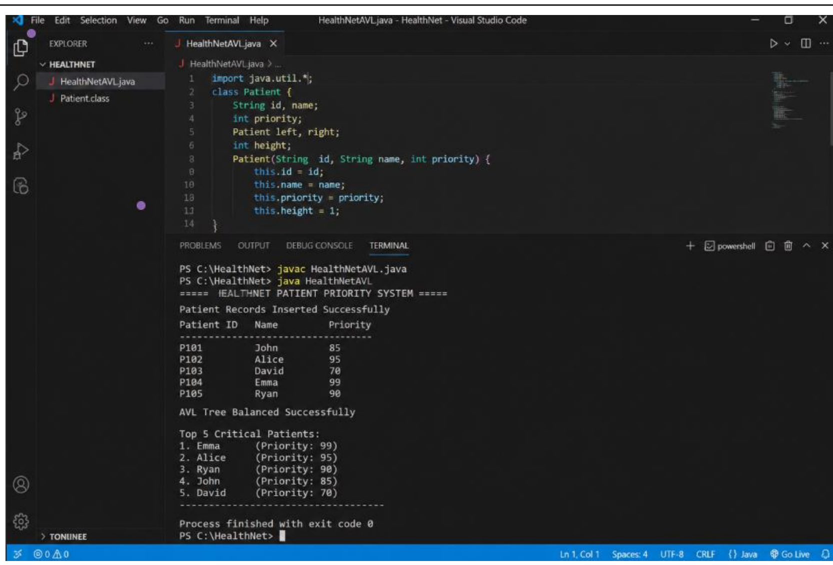
Work Done Summary:

In this case study, we implemented an **AVL Tree** for HealthNet's emergency patient management system.
The AVL Tree stores patient records according to their priority score (criticality level). Patients with higher priority can be accessed quickly during emergencies.

We inserted patient priorities into the AVL Tree while maintaining balance through AVL rotations.
A reverse in-order traversal was implemented to retrieve the highest-priority patients efficiently.

Implementation Details

1. Inserted patient priority scores into AVL Tree.
2. Maintained AVL balancing after every insertion.
3. Performed LL, RR, LR, and RL rotations when required.
4. Stored patient ID, name, and priority score.
5. Implemented reverse in-order traversal to retrieve critical patients.
6. Compared AVL Tree performance with a sorted ArrayList.
7. Evaluated time complexity for healthcare workloads.



Github link :

Student signature	Faculty signature